Chicago house prices
Assignment 5 summary

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What we did

We used the team draft to predict Chicago single-family sale prices from 2023-2025. We kept the outcome in dollars, cleaned the training rows once, and tested house features, predictor transformations, neighborhood characteristics and interactions. We used 10-fold cross-validation to compare 16 model trials.

We started with 23,200 training sales and retained 22,261 after removing 939 sales outside the log-price IQR bounds. We generated one finite prediction for each of the 5,789 test IDs, in the original order, without dropping any test rows.

Measure	Structure-only baseline	Final selected formula
CV RMSE	$219,610	$135,451
Residual Moran's I	0.468	0.169

The final formula reduces reported CV RMSE by 38.3% relative to the baseline. The lowest reported score in the log is m16, $135,451. The instructor-held-out test RMSE is still unavailable because the test prices are withheld.

How we evaluated the model

We calculated residual maps, Moran's I and valuation ratios from out-of-fold predictions. Numeric predictors are limited to the training range and predicted prices are floored at zero, with both safeguards applied inside CV and to test predictions. Crime counts exclude events on or after each sale. The income comparison examines whether relative valuation errors differ across neighborhoods.

Final files

Portfolio page: Open Assignment 5
Team file: team_injera_predictions.csv (id, pred).

Where the model fails

The largest community-level out-of-fold RMSE is in LOGAN SQUARE: $281,059, across 476 retained sales. Its mean prediction-minus-sale error is $-157. Positive error means overprediction; negative means underprediction. A high dollar RMSE can partly reflect a higher-price market.

Who faces overvaluation

Neighborhood income group	Sales	Median ratio	Overvalued
Lowest income	5,547	1.076	54.8%
Lower middle	5,547	1.025	52.8%
Upper middle	5,546	1.005	50.7%
Highest income	5,546	0.992	48.6%

A predicted-to-sale ratio above 1 means overvaluation relative to that sale. We compared retained sales in four groups using 2023 ACS five-year tract median household income. Income was used for this check only, not as a model predictor. This is not a direct calculation of household tax bills.

The median ratio is 1.076 in the lowest-income group and 0.992 in the highest-income group. This comparison matters because lower overall dollar error can still hide unequal relative errors, the concern raised in The Tax Divide.

Reading the model

Our location features include community effects, transit distances, nearby park facilities, earlier crime events, affordable-housing measures and ADU-area membership. Size-by-community and other interactions allow some house characteristics to have different effects in different settings. These are associations, not causal claims.

The bus-distance coefficient is $53,778.14 per mile. Holding other terms constant, an additional 0.1 miles changes the fitted price by $5,377.81. The coefficient table in the assignment gives the complete fitted estimates.

Limits before assessment use

We would not use this directly to set tax bills. Stepwise selection preceded CV, so its score can be optimistic; repeated sales can cross CV folds. The IQR rule removes unusual prices without proving they are non-market and can underrepresent low-value homes. Early 2023 has less crime history, and the supplied amenity files lack historical snapshot dates. Their compliance with the 2023-2025 source window cannot be certified from those files.

The guarded linear model produces 5 zero-valued test predictions. The floor prevents negative prices, but these cases remain a prediction limitation and should be reviewed before operational use.

Sources: team-supplied sales and Chicago spatial files; U.S. Census Bureau, 2023 ACS five-year B19013_001 and tract geometry. Detailed model log, code, diagnostic maps, contributions and AI statement appear on the portfolio page.